

Q1. CASE STUDY 1

```
code
#include <stdio.h>
struct Project {
    char pName[50];
    char role[50];
    int duration;
};
union Status {
    char fullTime[20];
    char contract[10];
    char intern[10];
};
struct Employee {
    int id;
    char name[60];
    char dept[50];
    char desig[50];
    float salary;
    struct Project proj;
    union Status status;
};
struct Employee emp;
int main() {
    scanf("%d", &emp.id);
    scanf("%60s", emp.name);
    scanf("%50s", emp.dept);
    scanf("%50s", emp.desig);
    scanf("%f", &emp.salary);
    scanf("%50s", emp.proj.pName);
    scanf("%50s", emp.proj.role);
    scanf("%d", &emp.proj.duration);
    scanf("%10s", emp.status.fullTime);
    printf("Employee ID: %d\n", emp.id);
    printf("Name: %s\n", emp.name);
    printf("Department: %s\n", emp.dept);
    printf("Designation: %s\n", emp.desig);
    printf("Salary: %.2f\n", emp.salary);
    printf("Project Name: %s\n", emp.proj.pName);
    printf("Project Role: %s\n", emp.proj.role);
    printf("Project Duration: %d months\n", emp.proj.duration);
    printf("Employment Status: %s\n", emp.status.fullTime);
    return 0;
}
```

INPUT	101 John IT Developer 75000.50 CloudMigration lead 0 Full-time
OUTPUT	(no output)

Q2. CASE STUDY 2

CODE

```
#include <stdio.h>
struct Developer {
    char name[50];
};
struct Tester {
    char name[50];
};
union Status {
    char state[50];
};
struct Bug {
    int id;
    char desc[100];
    char severity[50];
    struct Developer dev;
    struct Tester test;
    union Status stat;
};
struct Bug b;
int main() {
    scanf("%d", &b.id);
    scanf("%99[^\n]", b.desc);
    scanf("%99[^\n]", b.severity);
    scanf("%99[^\n]", b.dev.name);
    scanf("%99[^\n]", b.test.name);
    scanf("%99[^\n]", b.stat.state);
    printf("Bug ID: %d\n", b.id);
    printf("Description: %s\n", b.desc);
    printf("Severity: %s\n", b.severity);
    printf("Developer: %s\n", b.dev.name);
    printf("Tester: %s\n", b.test.name);
    printf("Status: %s\n", b.stat.state);
    return 0;
}
```

INPUT

```
101
Login error on homepage
High
Alex
Sam
Open
```

OUTPUT

(no output)

Q3. CASE STUDY 3

CODE

```
#include <stdio.h>

struct Supplier {
    char name[50];
};

union Category {
    char type[50];
};

struct Product {
    int id;
    char name[50];
    float price;
    int stock;
    union Category cat;
    struct Supplier sup;
};

struct Product p;

int main() {
    if (scanf("%d", &p.id) != 1) return 0;
    scanf("%49[^\n]", p.name);
    scanf("%f", &p.price);
    scanf("%d", &p.stock);
    scanf("%49[^\n]", p.cat.type);
    scanf("%49[^\n]", p.sup.name);

    printf("Product ID: %d\n", p.id);
    printf("Product Name: %s\n", p.name);
    printf("Price: %.2f\n", p.price);
    printf("Stock Quantity: %d\n", p.stock);
    printf("Category: %s\n", p.cat.type);
    printf("Supplier Name: %s\n", p.sup.name);
    return 0;
}
```

INPUT

```
100
Laptop
55000.50
15
Electronics
TechCorp
```

OUTPUT

```
(no output)
```

Q4. CASE STUDY 4

CODE

```
#include <stdio.h>

struct Visit {
    char diagnosis[50];
    char doctor[50];
    float cost;
};

union Status {
    char stat[50];
};

struct Patient {
    int id;
    char name[50];
    int age;
    char bloodGroup[10];
    char history[50];
    struct Visit visit;
    union Status status;
};

struct Patient p;

int main() {
    if (scanf("%d", &p.id) != 1) return 0;
    scanf("%49s", p.name);
    scanf("%d", &p.age);
    scanf("%9s", p.bloodGroup);
    scanf("%49s", p.history);
    scanf("%49s", p.visit.diagnosis);
    scanf("%49s", p.visit.doctor);
    scanf("%f", &p.visit.cost);
    scanf("%49s", p.status.stat);

    printf("Patient ID: %d\n", p.id);
    printf("Patient Name: %s\n", p.name);
    printf("Age: %d\n", p.age);
    printf("Blood Group: %s\n", p.bloodGroup);
    printf("Medical History: %s\n", p.history);
    printf("Diagnosis Details: %s\n", p.visit.diagnosis);
    printf("Doctor Name: %s\n", p.visit.doctor);
    printf("Treatment Cost: %.2f\n", p.visit.cost);
    printf("Patient Status: %s\n", p.status.stat);
    return 0;
}
```

INPUT

```
101
John Doe
45
O+
Hypertension
Viral Fever
Dr. Smith
1500.50
admitted
```

OUTPUT

(no output)